

Unit 2

SECTION I — Short Answers (2 Marks each)

Q15. Define the No-Cloning Theorem. [CO1, 2M]

The No-Cloning Theorem states that it is impossible to construct a general unitary operation that can create an identical, independent copy of an arbitrary, unknown quantum state.

Formally, no unitary operator U and fixed 'blank' state $|s\rangle$ exist such that, for every possible input state $|\psi\rangle$, the machine outputs two copies of $|\psi\rangle$:

$$U |\psi\rangle |s\rangle = |\psi\rangle |\psi\rangle \quad (\text{impossible for arbitrary } |\psi\rangle)$$

This is a fundamental consequence of the linearity of quantum mechanics and has no classical analogue, since classical bits can always be read and rewritten freely.

Q16. What is Quantum Teleportation? [CO1, 2M]

Quantum Teleportation is a protocol that transfers an unknown quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from a sender (Alice) to a distant receiver (Bob) without physically moving the particle itself.

It uses a pre-shared entangled Bell pair between Alice and Bob as a resource, together with a local Bell-basis measurement by Alice and the transmission of two classical bits to Bob, who applies a corresponding correction operation to reconstruct $|\psi\rangle$ exactly. The original qubit's state is destroyed at Alice's end during measurement, so the No-Cloning Theorem is not violated.

Q17. Explain why an unknown quantum state cannot be cloned. [CO1, 2M]

An unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ cannot be measured to learn α and β , because measurement collapses the superposition into $|0\rangle$ or $|1\rangle$, irreversibly destroying the original amplitudes.

Since copying via measure-and-recreate is ruled out, the only alternative is a unitary (linear) cloning machine U . But linearity forces U to act term-by-term on a superposition, which produces an entangled state rather than two independent product copies (see Q22). Hence, no process — measurement-based or unitary — can copy an arbitrary unknown state.

Q18. Differentiate between classical copying and quantum cloning. [CO1, 2M]

Aspect	Classical Copying	Quantum Cloning
Basic operation	Read the bit value (0 or 1), write it to a new memory slot	Attempt to duplicate amplitudes α, β of $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$
Effect of reading	Reading a bit does not disturb it	Measuring a qubit collapses/destroys its superposition
Feasibility	Always possible — trivial operation	Impossible for arbitrary/unknown states (No-Cloning Theorem)
Special case	N/A	Possible only for known orthogonal basis states, e.g. $ 0\rangle, 1\rangle$

Aspect	Classical Copying	Quantum Cloning
Underlying reason	Classical information is not disturbed by observation	Quantum linearity forces entanglement instead of independent copies

Q19. Explain the significance of Bell's Inequality. [CO1, 2M]

Bell's Inequality sets a strict mathematical upper limit ($|C| \leq 2$) on how strongly correlated the measurement outcomes of two distant particles can be if nature obeys classical physics with local hidden variables (local realism).

Its significance lies in being experimentally testable: quantum mechanics predicts correlations (up to $2\sqrt{2} \approx 2.828$, the Tsirelson bound) that clearly exceed this classical limit. Experimental violation of the inequality rules out local hidden-variable theories, confirms genuine quantum entanglement, and establishes the non-local character of quantum reality — forming the theoretical foundation for quantum cryptography and quantum information protocols.

Q20. Illustrate how entanglement is utilized in Quantum Teleportation. [CO1, 2M]

Entanglement is the essential resource that allows teleportation to bypass the No-Cloning Theorem. Alice and Bob first share a maximally entangled Bell pair:

$$|\beta_{00}\rangle_{23} = (1/\sqrt{2}) (|00\rangle + |11\rangle)$$

Alice's Bell-basis measurement on her qubit (the unknown state) and her half of the entangled pair instantaneously correlates with Bob's half, projecting Bob's qubit into a state related to $|\psi\rangle$ by a known Pauli operation (I, X, Z, or ZX).

Alice then sends the 2-bit measurement outcome through a classical channel; Bob applies the corresponding correction gate to recover $|\psi\rangle$ exactly. Without the shared entanglement, no classical channel alone could reconstruct the exact quantum state.

SECTION II — Long Answers (6 Marks each)

Q21. Explain the No-Cloning Theorem and derive its mathematical proof using the inner product preservation property of unitary operators. [CO2, 6M]

Concept

In classical computing, copying data is trivial — a bit (0 or 1) is read and rewritten to a new slot without disturbance. In quantum mechanics, if we have an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we cannot measure it to determine α and β , because measurement collapses the state. A cloning machine must therefore work by unitary evolution alone.

Step 1 — Define the cloning machine

A quantum copying machine is modeled as a unitary operator U acting on two inputs: the target state $|\psi\rangle$ to be copied, and a blank 'quantum paper' state $|s\rangle$. For the machine to work, it must output two copies of $|\psi\rangle$:

$$U |\psi\rangle|s\rangle = |\psi\rangle|\psi\rangle \quad \dots(1)$$

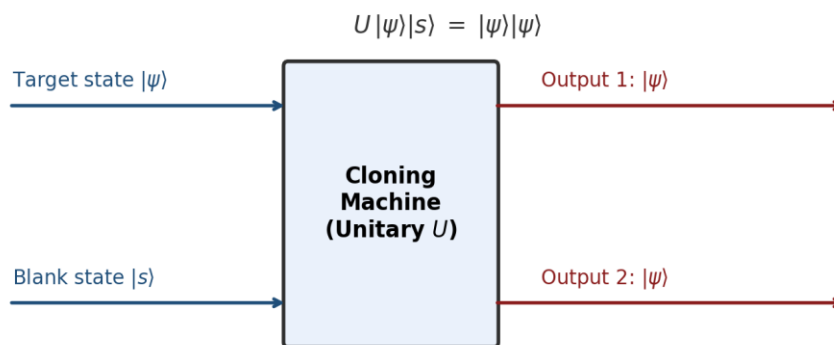


Fig 1: Schematic of a hypothetical universal quantum cloning machine.

Step 2 — Test with two different states

If U is a truly universal copier, it must work for any state. Suppose it successfully clones both $|\psi\rangle$ and a different state $|\phi\rangle$:

$$U |\psi\rangle|s\rangle = |\psi\rangle|\psi\rangle \quad \text{and} \quad U |\phi\rangle|s\rangle = |\phi\rangle|\phi\rangle \quad \dots(2)$$

Step 3 — Take the inner product

Compute the inner product of the states after cloning and equate it to the inner product before cloning, since unitary operators preserve inner products.

(a) After cloning — inner product of the right-hand sides of (2):

$$(\langle\psi|\langle\psi|)(|\phi\rangle|\phi\rangle) = \langle\psi|\phi\rangle\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2$$

(b) Before cloning — inner product of the left-hand sides (before U acts). Since U is unitary, $U^\dagger U = I$, so it preserves inner products and drops out completely:

$$(\langle\psi|\langle s|U^\dagger)(U|\phi\rangle|s\rangle) = \langle\psi|\langle s| |\phi\rangle|s\rangle = \langle\psi|\phi\rangle\langle s|s\rangle$$

Since the blank state $|s\rangle$ is normalized, $\langle s|s\rangle = 1$. This leaves $\langle\psi|\phi\rangle$.

Step 4 — The conflict

Because a unitary operation cannot change the inner product, the "before" value must equal the "after" value:

$$(\langle \psi | \phi \rangle)^2 = \langle \psi | \phi \rangle$$

Let $x = \langle \psi | \phi \rangle$. This gives the algebraic equation:

$$x^2 = x \Rightarrow x^2 - x = 0 \Rightarrow x(x - 1) = 0$$

This has exactly two solutions:

- If $x = 1$: $\langle \psi | \phi \rangle = 1$, so $|\psi\rangle = |\phi\rangle$ (the states are identical).
- If $x = 0$: $\langle \psi | \phi \rangle = 0$, so the states are perfectly orthogonal (like $|0\rangle$ and $|1\rangle$).

Conclusion

A cloning machine can be built to perfectly copy orthogonal basis states such as $|0\rangle$ and $|1\rangle$. But if two states have any overlap between 0 and 1 (as in a superposition state), the mathematics breaks down — no consistent unitary U can satisfy the cloning equation. Therefore, a universal quantum cloning machine cannot exist.

Q22. Describe the cloning of basis states and superposition states. Explain why cloning fails for arbitrary states with suitable mathematical analysis. [CO2, 6M]

Cloning of basis states

Suppose a device perfectly clones the standard basis states $|0\rangle$ and $|1\rangle$:

$$U|0\rangle|s\rangle = |0\rangle|0\rangle \quad U|1\rangle|s\rangle = |1\rangle|1\rangle$$

This is perfectly consistent with unitarity (basis states are orthogonal, so $\langle 0|1\rangle = 0$ satisfies $x(x-1)=0$ from Q21).

Attempting to clone a superposition state

Now pass the diagonal superposition state $|+\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$ through the same machine.

Path 1 — What a perfect clone should look like: if cloning worked, the output should be two independent copies of $|+\rangle$:

$$\text{Expected} = |+\rangle|+\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle) \otimes (1/\sqrt{2})(|0\rangle + |1\rangle)$$

$$\text{Expected} = (1/2)(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Path 2 — What quantum mechanics actually outputs (by linearity): since QM is strictly linear, U must distribute across the superposition term-by-term:

$$U|+\rangle|s\rangle = U[(1/\sqrt{2})(|0\rangle + |1\rangle)]|s\rangle = (1/\sqrt{2})(U|0\rangle|s\rangle + U|1\rangle|s\rangle)$$

Substituting the known machine behaviour $U|0\rangle|s\rangle = |00\rangle$ and $U|1\rangle|s\rangle = |11\rangle$:

$$\text{Actual output} = (1/\sqrt{2})(|00\rangle + |11\rangle)$$

Verdict — comparison

Quantity	Expression
Expected (independent copies)	$(1/2)(00\rangle + 01\rangle + 10\rangle + 11\rangle)$
Actual (linear machine output)	$(1/\sqrt{2})(00\rangle + 11\rangle)$

Instead of creating two independent copies of the state $|+\rangle|+\rangle$, the machine produced a highly entangled Bell state. The two qubits are bound together rather than cloned — entanglement forces the cloning process to fail.

General principle

- A CNOT-type gate designed as a copier works perfectly for standard classical basis states $|0\rangle$ and $|1\rangle$.
- If fed a superposition state, the machine entangles the two qubits together instead of making a copy.
- What was wanted: two completely independent, isolated qubits both in state $|\psi\rangle$ — i.e. $|\psi\rangle \otimes |\psi\rangle$.
- What is actually obtained: a single, highly entangled Bell-like state — modifying or measuring one qubit instantly affects the other.

Q23. Explain the Quantum Teleportation protocol with the quantum circuit and describe the roles of Alice and Bob. [CO2, 6M]

Alice (sender) and Bob (receiver) play the major roles in quantum teleportation:

- Alice: holds a qubit in an unknown, arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and wants to send this exact state to Bob.
- Bob: is space-separated from Alice, and waits to reconstruct $|\psi\rangle$ using information Alice sends him.

Due to the No-Cloning Theorem, Alice cannot make a perfect copy of an unknown quantum state without destroying the original — if she measures $|\psi\rangle$ directly, it collapses to $|0\rangle$ or $|1\rangle$ and she permanently loses the probability amplitudes α and β . To bypass this, quantum teleportation uses quantum entanglement as a resource.

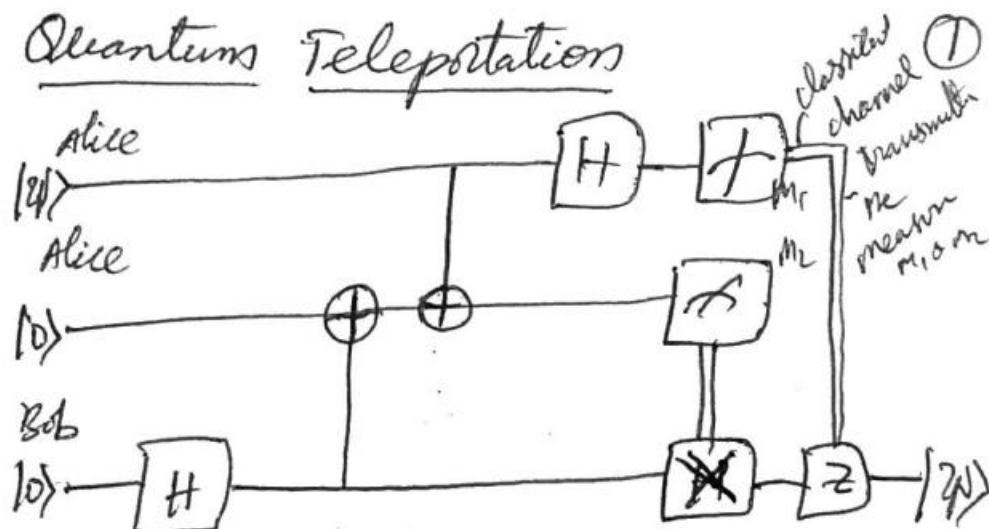
Resources required

- The target state to be teleported (held by Alice): $|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle$
- A shared maximally entangled Bell pair distributed between Alice (qubit 2) and Bob (qubit 3): $|\beta_{00}\rangle_{23} = (1/\sqrt{2})(|00\rangle + |11\rangle)$

Protocol steps (roles)

- Step 1: Alice applies a CNOT gate (qubit 1 $\rightarrow$ qubit 2) followed by a Hadamard gate on qubit 1, rotating her two qubits into the Bell basis.
- Step 2: Alice measures both her qubits, obtaining two classical bits m_1, m_2 , and destroying her local copy of $|\psi\rangle$ (consistent with No-Cloning).
- Step 3: Alice sends m_1, m_2 to Bob over an ordinary classical channel.
- Step 4: Bob applies a correction — an X gate if $m_2 = 1$, and a Z gate if $m_1 = 1$ — to his qubit, exactly reconstructing $|\psi\rangle$.

Quantum Teleportation



Here Alice (Sender) & Bob (Receiver) plays a major role. where

Alice: Has a qubit in an unknown arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

she wants to send his exact state to Bob

Bob: Space-separated from Alice, waiting to reconstruct $|\psi\rangle$

Due to no cloning theorem, Alice cannot make a perfect copy of an unknown quantum state without destroying the original. If Alice measures $|\psi\rangle$, the state collapses to $|0\rangle$ or $|1\rangle$ and she permanently loses the probability amplitudes α & β .

TO bypass this, quantum teleportation ^②
uses quantum entanglement as a resource.
TO teleport one qubit we require

1. The target state to be teleported (held by Alice)

$$|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle$$

2. A shared Maximally Entangled pair
(Bell state) distributed b/w Alice & Bob

$$|\Phi_{00}\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Qubit 2 belongs to Alice

Qubit 3 belongs to Bob

$\therefore$ Total initial state i.e., combined state
of entire 3-qubit system (Qubit 1 & 2 with
Alice & 3 with Bob) is a tensor product

$$|\psi_0\rangle = |\psi\rangle_1 \otimes |\Phi_{00}\rangle_{23}$$

$$= (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Expanding mathematically gives

(3)

$$|2\psi_0\rangle = \frac{1}{\sqrt{2}} [\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)]$$

$$|2\psi_0\rangle = \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]$$

Now step by step protocol execution

Step 1: Alice needs to change the basis of her two qubits (1 & 2) into the Bell Basis she does this using 2 quantum gates

1- Apply CNOT Gate with Qubit 1 as control and Qubit 2 as target (CNOT₁₂)

• If qubit 1 is $|0\rangle$, qubit 2 stays the same

• If qubit 1 is $|1\rangle$, qubit 2 flips

$$|2\psi_1\rangle = \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]$$

2. Apply a Hadamard gate to Qubit 1 (H_1)⁴

$$\text{Remember } H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Substitute the into $|\Psi_1\rangle$.

$$|\Psi_2\rangle = \frac{1}{2} [\alpha(|0\rangle + |1\rangle)|00\rangle + \alpha(|0\rangle + |1\rangle)|11\rangle \\ + \beta(|0\rangle - |1\rangle)|10\rangle + \beta(|0\rangle - |1\rangle)|01\rangle]$$

Step 2: If we expand the expressions above and regroup the terms based on Alice's qubit

$$|\Psi_2\rangle = \frac{1}{2} [\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle \\ + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \\ \beta|001\rangle - \beta|101\rangle]$$

1) Grouping terms starting with $|00\rangle$

$$\alpha|000\rangle + \beta|001\rangle$$

$$\Rightarrow |00\rangle(\alpha|0\rangle + \beta|1\rangle)$$

2) $|01\rangle$

$$\alpha|011\rangle + \beta|010\rangle \Rightarrow |01\rangle(\alpha|1\rangle + \beta|0\rangle)$$

3) $|10\rangle$

(5)

$$\alpha|100\rangle - \beta|101\rangle \Rightarrow |10\rangle (\alpha|0\rangle - \beta|1\rangle)$$

4) $|11\rangle$

$$\alpha|111\rangle - \beta|110\rangle \Rightarrow |11\rangle (\alpha|1\rangle - \beta|0\rangle)$$

$\therefore$

$$|2N_2\rangle = \frac{1}{2} \left[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) \right. \\ \left. + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle) \right]$$

$\therefore$ If

Alice has $|00\rangle$, Bob has: $\alpha|0\rangle + \beta|1\rangle$.
(Exact original state)

Alice has $|01\rangle$, Bob has $\alpha|1\rangle + \beta|0\rangle$
(Flipped state)

Alice has $|10\rangle$, Bob has: $\alpha|0\rangle - \beta|1\rangle$
(Phase shift state)

Alice has $|11\rangle$, Bob has $\alpha|1\rangle - \beta|0\rangle$
(Flipped & phase shift state)

Q24. Derive the mathematical steps involved in the Quantum Teleportation algorithm and explain the state transformation after each quantum operation. [CO2, 6M]

Similar answer same as Q23

Q25. Explain Bell's Inequality, derive the CHSH inequality, and discuss its significance in quantum mechanics. [CO2, 6M]

Concept

Bell's inequality is a mathematical framework that sets a strict upper limit on how strongly correlated the measurement outcomes of two distant particles can be under the laws of classical physics (local realism).

Any physical system obeying classical mechanics cannot exceed this boundary, while quantum mechanics predicts a clear violation of it. Albert Einstein and colleagues (the EPR argument) proposed that quantum mechanics was incomplete, and that particles must possess predetermined hidden properties called Local Hidden Variables (LHVs) that dictate measurement outcomes before they occur. Bell proved theoretically that if such hidden variables existed, the correlation coefficient in a specific experimental setup must strictly remain ≤ 2 .

CHSH / Bell Test Setup

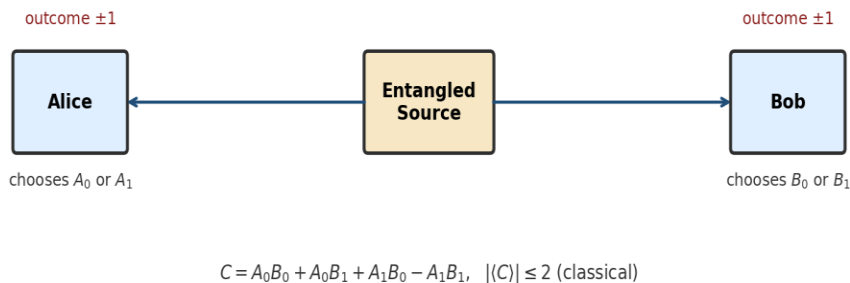


Fig 3: CHSH / Bell test setup — an entangled source sends one particle to Alice and one to Bob.

Mathematical derivation (CHSH form)

Setup: An entanglement source generates pairs of photons — one photon goes to Alice, the other to Bob.

- Alice can choose to measure her photon along two different angles: A_0 or A_1
- Bob can choose to measure his photon along two different angles: B_0 or B_1
- The outcomes of these measurements are binary: +1 (spin up) or -1 (spin down)

Assume local realism is true: outcomes A_0, A_1, B_0, B_1 all have predetermined values of either +1 or -1 simultaneously. Consider the algebraic combination:

$$\mathbf{C = A_0(B_0+B_1) + A_1(B_0-B_1)}$$

Rearranging:

$$\mathbf{C = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1}$$

Because B_0 and B_1 are each either +1 or -1, look at the terms (B_0+B_1) and (B_0-B_1) :

- If $B_0 = B_1$: then $(B_0+B_1) = \pm 2$ and $(B_0-B_1) = 0$
- If $B_0 \neq B_1$: then $(B_0+B_1) = 0$ and $(B_0-B_1) = \pm 2$

In either case, since A_0 and A_1 are also ± 1 , the value of C for any single pair of particles must strictly be either +2 or -2. Averaging over many trials gives Bell's (CHSH) Inequality:

$$| \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle | \leq 2$$

This value 2 is the absolute limit of correlation dictated by classical physics and local hidden variables.

Significance

- Rejection of local hidden-variable theories
- Confirmation of quantum entanglement
- Establishes the non-local nature of reality
- Major impact on the concept of classical realism
- Provides a foundation for quantum computing
- Enables applications in quantum cryptography
- Confirmed repeatedly through experimental verification

Q26. Derive the mathematical proof showing the violation of Bell's Inequality by quantum mechanics using an entangled Bell state. [CO2, 6M]

Let us see what quantum mechanics actually allows. If Alice and Bob share a maximally entangled Bell state, e.g.

$$|\psi^-\rangle = (1/\sqrt{2})(|01\rangle + |10\rangle)$$

and carefully choose their measurement angles, the quantum expectation values yield:

$$\langle A_0 B_0 \rangle = 1/\sqrt{2}, \quad \langle A_0 B_1 \rangle = 1/\sqrt{2}, \quad \langle A_1 B_0 \rangle = 1/\sqrt{2}, \quad \langle A_1 B_1 \rangle = -1/\sqrt{2}$$

Plugging these quantum values into the Bell (CHSH) inequality expression:

$$| 1/\sqrt{2} + 1/\sqrt{2} + 1/\sqrt{2} - (-1/\sqrt{2}) | = 4/\sqrt{2} = 2\sqrt{2}$$

$$2\sqrt{2} \approx 2.828$$

Since $2.828 > 2$, quantum mechanics explicitly violates Bell's inequality. This value $2\sqrt{2}$ is known as the Tsirelson bound — the maximum correlation allowed by quantum mechanics, strictly greater than the classical bound of 2, and it has been confirmed by numerous real-world Bell-test experiments (e.g. Aspect's experiments and later loophole-free tests).

This proves that no local hidden-variable theory can reproduce the predictions of quantum mechanics — the correlations between entangled particles are genuinely non-classical.

Q27. Explain the implications of the No-Cloning Theorem and discuss its role in Quantum Communication and Quantum Cryptography. [CO2, 6M]

Core implication

The No-Cloning Theorem proves that an arbitrary unknown quantum state cannot be perfectly duplicated by any physical process — unitary evolution or measurement. Formally, no U and $|s\rangle$ satisfy $U|\psi\rangle|s\rangle = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$ (derivation in Q21). This has deep implications across quantum information science.

Implications for Quantum Communication

- Any eavesdropper who intercepts a qubit in transit cannot make a perfect copy to analyze while forwarding an identical, undisturbed original — attempting to measure or clone it inevitably disturbs the state.
- This underlies quantum teleportation itself: since a state cannot be copied, information transfer must occur by destructively transferring (teleporting) it using entanglement plus classical communication, rather than duplicating it (Q23–Q24).
- It also justifies why quantum channels cannot be 'amplified' or repeated the way classical optical/electronic signals are — direct amplification would require cloning, which is forbidden.

Implications for Quantum Cryptography

- Quantum Key Distribution (QKD) protocols, such as BB84, rely directly on the No-Cloning Theorem for their security guarantee.
- An eavesdropper (Eve) intercepting qubits encoding a secret key cannot clone them to keep a copy for herself while sending on an unmodified original to the receiver.
- Any attempt by Eve to measure/clone the qubits introduces detectable disturbances (increased error rate) in the transmitted key, alerting the legitimate parties (Alice and Bob) to the presence of eavesdropping.
- This gives quantum cryptography a form of security that is guaranteed by the laws of physics, rather than by computational hardness assumptions used in classical cryptography.

Summary

The No-Cloning Theorem is not a limitation but a foundational security feature — it guarantees that quantum information cannot be silently intercepted and copied, forming the physical basis for secure quantum communication and cryptographic protocols.

Q28. Analyze the relationship among the No-Cloning Theorem, Quantum Teleportation, and Bell's Inequality. Justify how these principles establish the foundations of secure quantum communication. [CO2, 6M]

How the three principles connect

Principle	What it establishes	Role in the overall system
No-Cloning Theorem	An unknown quantum state cannot be perfectly copied (Q21)	Guarantees that quantum information cannot be secretly intercepted/duplicated — forces information transfer to be destructive
Quantum Teleportation	A state can still be transferred (not copied) using entanglement + 2 classical bits (Q23–Q24)	Provides the working protocol that respects No-Cloning while still moving quantum information between distant parties
Bell's Inequality	Entanglement correlations exceed the classical limit of 2, up to $2\sqrt{2}$ (Q25–Q26)	Proves the entangled resource used in teleportation is genuinely non-classical and non-local, not explainable by hidden variables

Logical chain

- Because the No-Cloning Theorem forbids copying, secure transfer of quantum information cannot rely on duplication — it must rely on a resource that is consumed in the process.
- Quantum teleportation supplies exactly this: it consumes a shared entangled Bell pair and 2 classical bits to move $|\psi\rangle$ from Alice to Bob, while destroying Alice's original copy — so at no point does more than one copy of $|\psi\rangle$ exist, consistent with No-Cloning.
- Bell's inequality violation experimentally certifies that the entanglement used in teleportation (and in QKD protocols) is real, non-local quantum correlation — not a classical hidden-variable trick that an eavesdropper could exploit or replicate.

Justification: foundations of secure quantum communication

Together these three results form a self-consistent physical foundation for security:

- No-Cloning $\Rightarrow$ an eavesdropper cannot copy transmitted quantum states without detection, and information cannot be silently duplicated.
- Teleportation $\Rightarrow$ demonstrates a concrete, physically realizable protocol for transferring quantum information using entanglement, obeying the No-Cloning restriction.
- Bell's Inequality violation $\Rightarrow$ certifies that the entanglement resource is genuinely quantum, ruling out any classical local-hidden-variable explanation an adversary could exploit to fake or predict the correlations.

Consequently, protocols such as Quantum Key Distribution (BB84, E91) and quantum teleportation-based networks derive their security directly from these three linked principles: unclonability prevents covert copying, teleportation shows how information moves without duplication, and Bell violation certifies the non-classical, tamper-evident nature of the shared entanglement — together establishing the physical foundations of secure quantum communication.